

 **Red Hat**

MULTICLUSTER VM GOVERNANCE

ACM Virtual Machine Management

*Red Hat Advanced Cluster Management + OpenShift
Virtualization*

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Why This Workshop Exists

The enterprise VM governance gap.

THE PROBLEM

Managing 200+ VMs across 15 clusters is broken

- No centralized VM governance — each cluster team manages policies independently, causing configuration drift
- Fragmented observability — operators log into each cluster individually, wasting hours per incident
- Overprivileged access — developers have cluster-admin everywhere because "that is how it has always been done"
- Destruction risk — a single `oc delete managedcluster` can take down production across an entire region
- Manual deployment — VM provisioning requires direct CLI access and tribal knowledge

A unified Kubernetes-native VM control plane

- Organizations are migrating away from legacy hypervisor monopolies — but many workloads cannot yet be containerized
- OpenShift Virtualization runs KVM-based VMs natively on OpenShift — same platform, same tooling
- RHACM provides the governance layer — policies, RBAC, observability, and lifecycle management at fleet scale
- Result: one control plane for both containers and VMs across every cluster in every region

From fragmented VM management to unified multicluster governance.



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Learning Objectives

Eight capabilities you will prove today.

EIGHT CAPABILITIES

What you will prove in this workshop

- 1 · Deploy VMs at fleet scale via GitOps
- 2 · Visualize cross-cluster application topology
- 3 · Enforce VM governance as code
- 4 · Achieve fleet-wide observability
- 5 · Right-size VMs with built-in recommendations
- 6 · Deploy VMs without cluster-admin
- 7 · Implement fine-grained RBAC with MRA
- 8 · Eradicate cluster destruction entirely

Each module uses a Know (business context) and Show (hands-on) format.



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Technology Stack

Products, operators, and APIs.

Core platform components

RHACM 2.17+	Advanced Cluster Management	Multicluster governance, policy engine, fleet management, fine-grained RBAC
OCP 4.22+	OpenShift Container Platform	Kubernetes platform with enterprise security and developer tooling
CNV 4.22+	OpenShift Virtualization	KVM-based VM hosting — runs virtual machines natively on OpenShift
OADP 1.6+	API for Data Protection	Velero-based VM backup with kubevirt plugin for consistent snapshots
GitOps 1.21+	OpenShift GitOps (ArgoCD)	Declarative VM deployment via ApplicationSets across the fleet
Pipelines	OpenShift Pipelines (Tekton)	Windows VM golden image build pipelines — ISO to running VM

Governance components in action today

Policy Engine	ConfigurationPolicy	Declarative compliance — enforce VM config, backup, network, security policies
Observability	Grafana + Thanos	Fleet-wide metrics aggregation with S3-backed long-term storage
Fleet Virt	Virtualization Perspective	Centralized VM lifecycle — start, stop, restart, snapshot via cluster-proxy
Fine RBAC	MulticlusterRoleAssignment	Distribute scoped Kubernetes roles across managed clusters (RHACM 2.17)
Search	search-collector addon	Real-time metadata streaming from managed clusters to hub
Admission	ValidatingAdmissionPolicy	CEL-based cluster destruction prevention — defense-in-depth

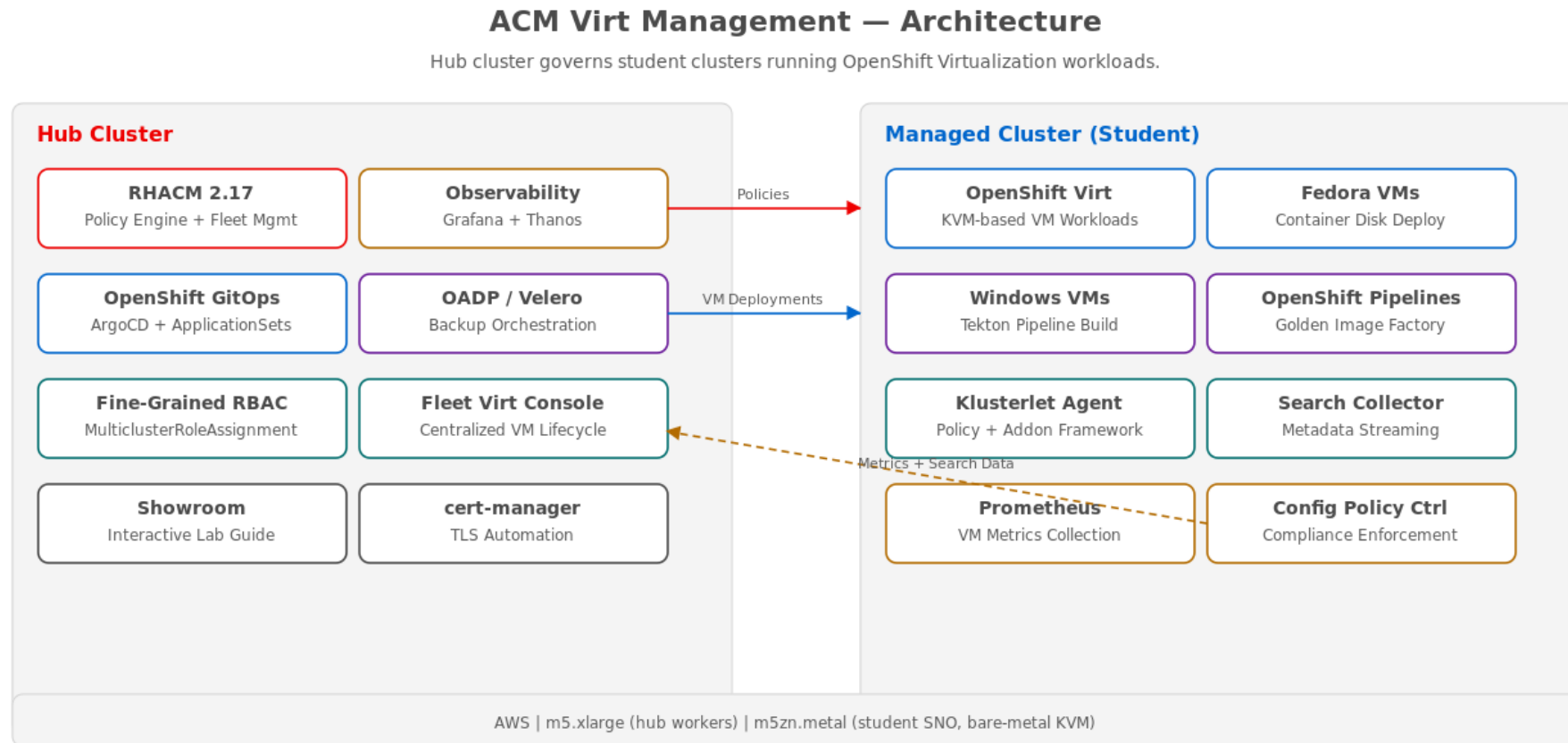


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Architecture

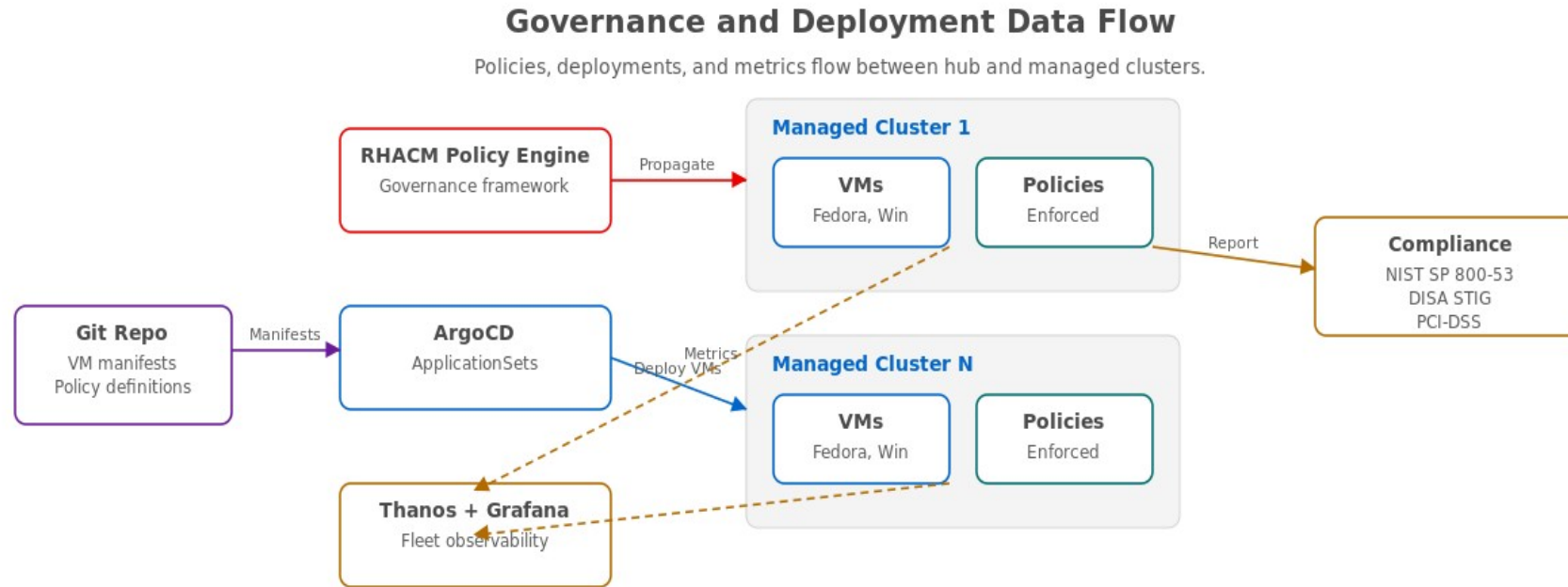
Hub and managed cluster topology.

Hub and managed cluster architecture



Hub runs RHACM, GitOps, and Observability. Managed clusters run OpenShift Virtualization workloads.

Governance, deployment, and metrics pipeline



Git-driven policies and VMs propagate to clusters; metrics and compliance data flow back to the hub.

A decorative vertical bar on the left side of the slide, featuring a collage of tech-related icons in various shades of red and orange. The icons include a cloud with a keyhole, a database cylinder, a server rack, a document with a magnifying glass, a gear, a lightbulb, a network diagram with nodes and arrows, and a computer monitor. The background of the slide is a solid dark red color.

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Workshop Agenda

Eight modules, approximately 90 minutes.

AGENDA

Workshop flow — eight modules

Module 1: Deploy VM Workloads

- *ArgoCD + Tekton — 10 min*

Module 2: Application Topology

- *Interactive graphs + Fleet Virt — 8 min*

Module 3: VM Policies and Governance

- *Compliance as code — 10 min*

Module 4: Fleet Observability

- *Grafana + Thanos dashboards — 10 min*

Module 5: VM Right-Sizing

- *Replace VMware DRS — 15 min*

Module 6: Deploy Without Cluster-Admin

- *Git Subscriptions — 8 min*

Module 7: Fine-Grained Permissions

- *MulticloudRoleAssignment — 10 min*

Module 8: Eradicate Cluster Destruction

- *Defense-in-depth — 6 min*

Total workshop time: ~90 minutes. Each module builds on the previous.



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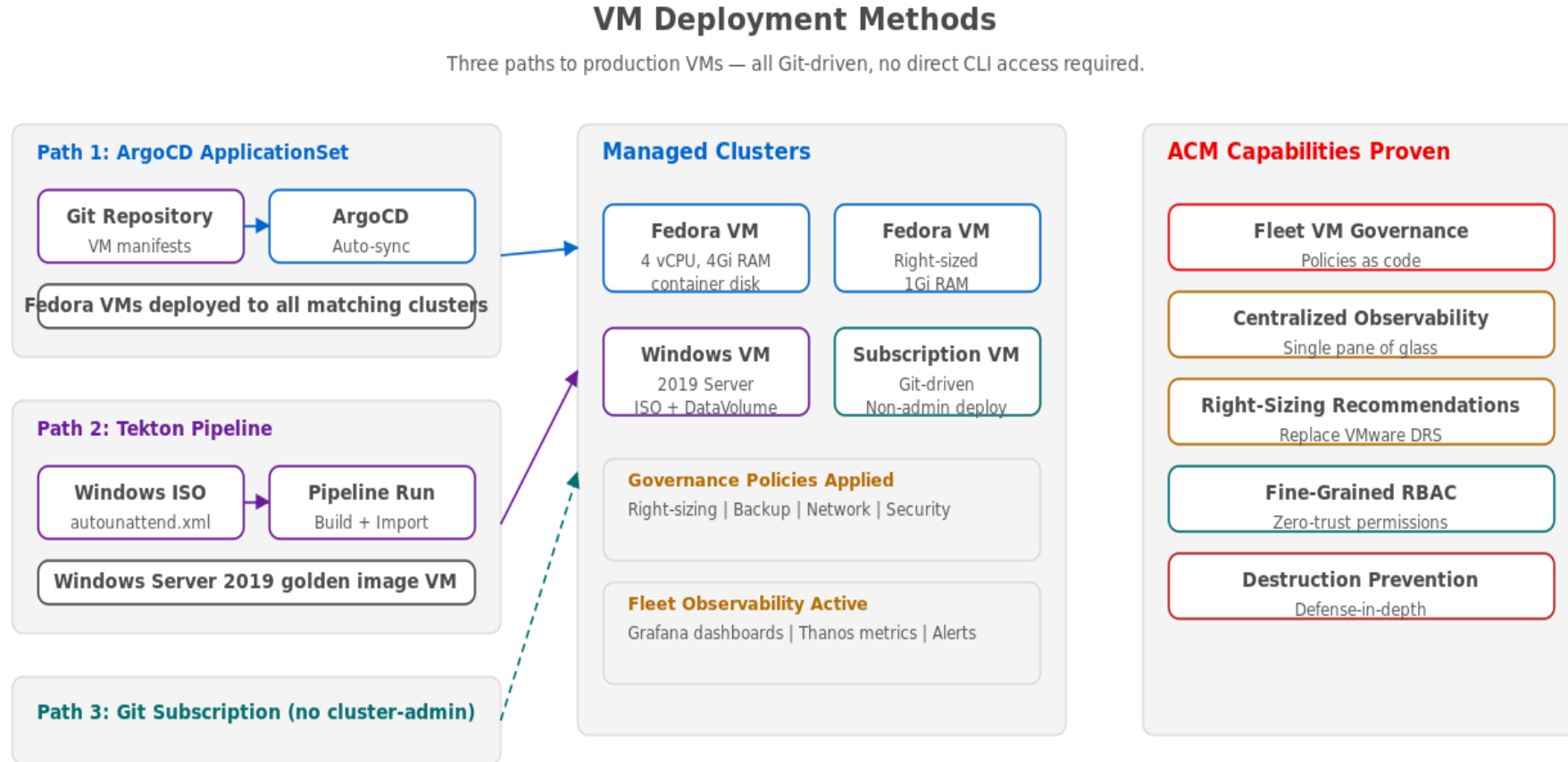
Module Deep Dives

Key concepts from each module.

Deploy VM workloads at fleet scale

- ArgoCD ApplicationSets with clusterDecisionResource generator — new clusters automatically receive VMs
- Fedora VMs deployed from container disks (quay.io/containerdisks/fedora) — no ISO management needed
- Windows Server 2019 built via Tekton pipeline — ISO import, autounattend.xml, DataVolume creation
- ManagedClusterSet + Placement + GitOpsCluster bind ArgoCD to RHACM fleet topology
- Git is the single source of truth — no human touches a cluster CLI for deployment

Three paths to production VMs



ArgoCD for Linux fleet, Tekton for Windows golden images, Git Subscriptions for non-admin deployments.

Application topology and Fleet Virtualization

- ACM topology graph shows cross-cluster dependencies — Subscriptions, Deployments, Pods, VMs in one interactive view
- Fleet Virtualization perspective provides centralized VM lifecycle management (RHACM 2.16+)
- Start, stop, restart, pause, snapshot VMs across all clusters from the hub console
- VM operations proxy through the cluster-proxy framework — no direct cluster access needed
- Remote log retrieval for VMs directly from the topology view

VM policies and governance as code

Right-Sizing	PrometheusRule	Recording rules for CPU and memory utilization analysis (NIST CM-2)
Backup	OADP + Velero	Daily backup schedules enforced across all VM namespaces (NIST CP-9)
Network	NetworkPolicy	Default-deny ingress — explicit allow required per service (NIST SC-7)
Resources	LimitRange	CPU/memory guardrails prevent resource contention (NIST CM-7)
Eviction	VirtualMachinePreference	Eviction strategy ensures graceful VM handling (NIST CP-2)
Security	ClusterPreference	Hardened boot and security configuration baselines (NIST SC-28)

All policies mapped to NIST SP 800-53 controls — also applicable to DISA STIG, HIPAA, and PCI-DSS.

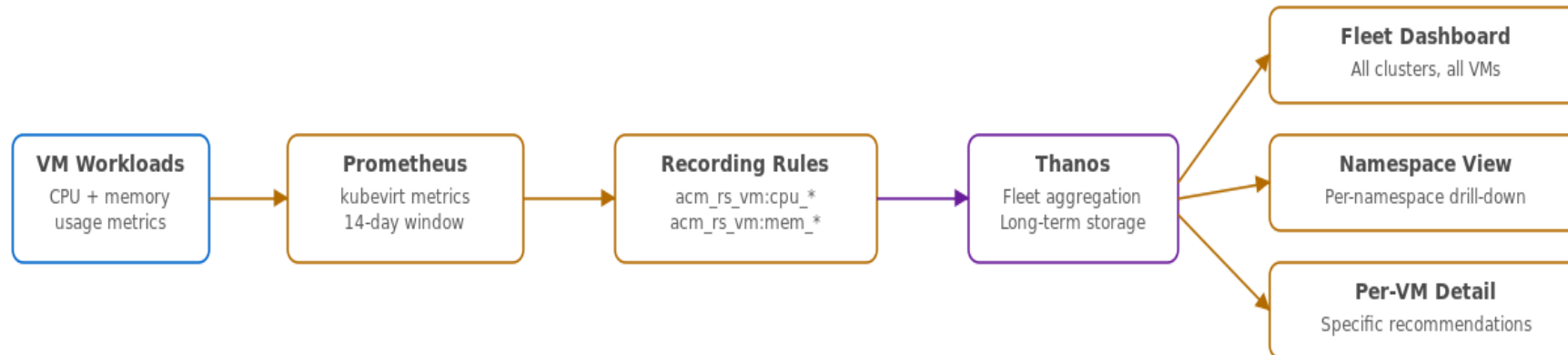
Fleet-wide observability with custom dashboards

- Cluster Overview dashboard — fleet-wide health at a glance across all managed clusters
- Single cluster deep dive — per-cluster resource utilization, VM density, and health metrics
- Custom VM fleet dashboard deployed as code — dashboards-as-code via ConfigurationPolicy
- Thanos Ruler for custom alerting — trigger on fleet-wide VM conditions, not just per-cluster
- Right-sizing opportunity identification — surface VMs consuming more resources than needed

VM right-sizing — replace VMware DRS

VM Right-Sizing Pipeline

Kubernetes-native replacement for VMware DRS — continuous, automated recommendations.



Action: Operator applies live resize based on recommendation

Example: Fedora VM resized from 4Gi to 1Gi RAM — zero downtime, verified in Grafana

Thresholds configurable via rs-virt-config ConfigMap — tailor to your SLAs

Continuous right-sizing recommendations with live resize capability — zero downtime.

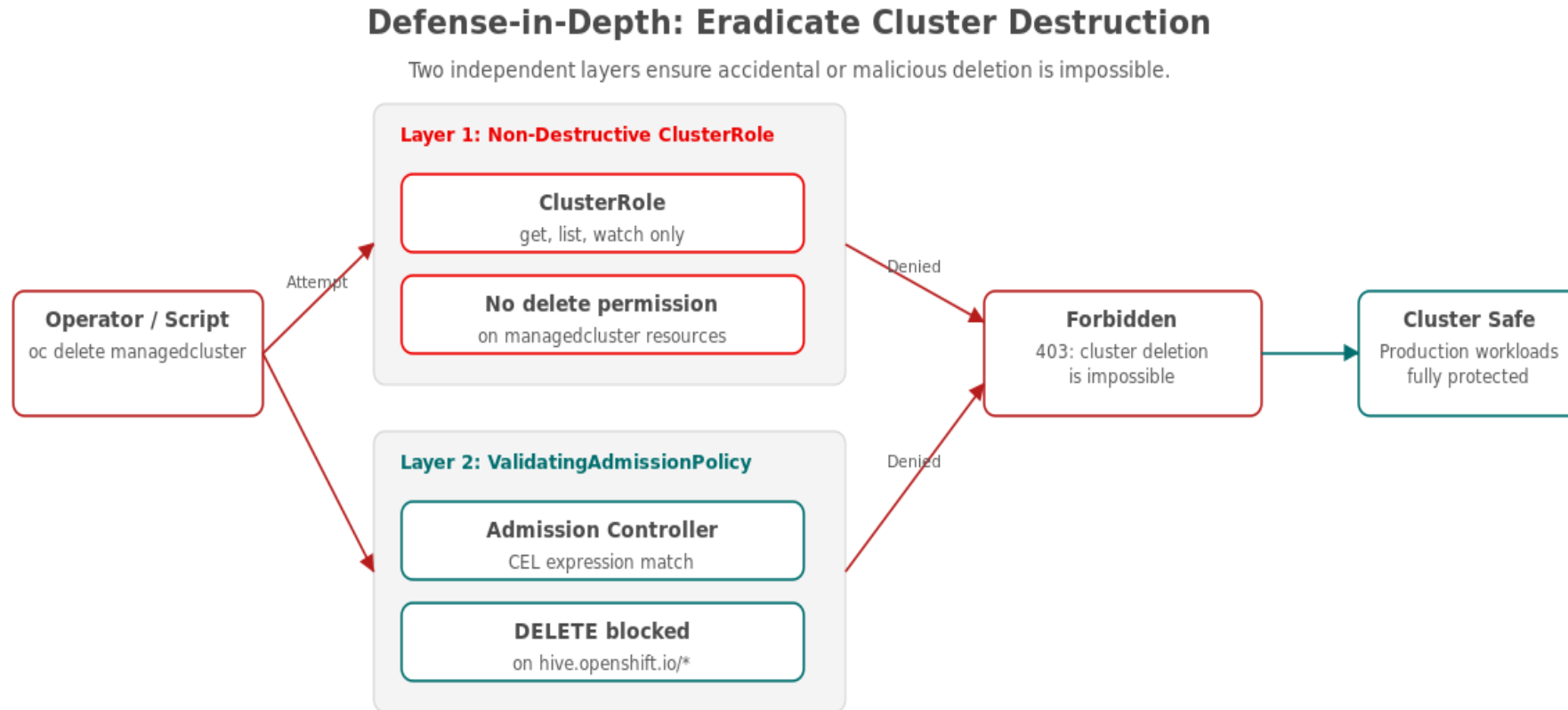
Deploy VMs without cluster-admin

- The problem: traditional VM deployment requires cluster-admin — an overprivileged, risky practice
- Solution: Git Subscription model — Channel + Placement + Subscription resources
- The subscription-admin ClusterRoleBinding grants deploy rights without cluster-admin privileges
- VM manifests live in Git, RHACM propagates them — operators never need direct cluster access
- Verification: prove that the non-admin user cannot perform destructive operations

Fine-grained permissions with MulticlusterRoleAssignment

- Enable fine-grained RBAC in MultiClusterHub — a foundational capability in RHACM 2.17
- Two-level model: hub-level roles control ACM console access, spoke-level roles control cluster actions
- MulticlusterRoleAssignment (MRA) distributes kubevirt.io:edit role to specific users on specific clusters
- Console-based role assignment UI for day-two operations — no YAML required
- Demo: compare admin console view vs. scoped vm-operator console — dramatic difference in permissions

Eradicate cluster destruction — defense-in-depth



Two independent safety layers make accidental or malicious cluster deletion architecturally impossible.



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Expected Outcomes

What you will walk away with.

What this workshop demonstrates

- Unified governance — one policy engine managing VMs and containers across all clusters
- Operational efficiency — fleet-wide observability eliminates per-cluster context switching
- Zero-trust deployment — Git is the single source of truth, no human CLI access to production
- Cost optimization — right-sizing recommendations surface overprovisioned VMs for immediate savings
- Compliance readiness — policies mapped to NIST SP 800-53, DISA STIG, HIPAA, and PCI-DSS
- Architectural safety — cluster destruction is impossible, not just discouraged

NEXT STEPS

Recommended path forward

- 1 · Proof of Concept — deploy RHACM + OpenShift Virtualization in your environment
- 2 · Policy Customization — tailor governance policies to your compliance requirements and SLAs
- 3 · RBAC Design Workshop — map your team structure to MulticloudRoleAssignment configurations
- 4 · Migration Planning — use Migration Toolkit for Virtualization (MTV) for existing VM estates
- 5 · Progressive Rollout — start with non-production clusters, expand as confidence grows

Your Red Hat account team can help scope and plan each of these steps.



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Questions?

Let us discuss.

THANK YOU

Ready for the hands-on workshop

- Workshop guide available in your Showroom environment
- Each module includes click-to-execute commands — no copy-paste errors
- Presenter: Tosin Akinosho
- All workshop content is open source and available for reference

Let's begin the hands-on exercises.